

EGERTON



UNIVERSITY

UNIVERSITY EXAMINATIONS

REGULAR -NJORO CAMPUS

SECOND SEMESTER, 2020/2021 ACADEMIC YEAR

THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN

ENVIRONMENTAL SCIENCE

ENSC 343: ENVIRONMENTAL POLLUTION AND CONTROL

STREAM: BSc (ENSC)

TIME: 2 HRS

EXAMINATION SESSION: DECEMBER

YEAR: 2021

INSTRUCTIONS:

- (i) Attempt all questions in Section A and any TWO in Section B
- (ii) Do not write on the question paper
- (iii) Illustrate your answers with suitable diagrams and formulas where appropriate

SECTION A (40 MARKS)

Question One

Write short notes on the following terms.

- a) Control of noise pollution **(3 Marks)**
- b) Temperature inversion in a city **(3 Marks)**
- c) Classification of pollutants **(3 Marks)**

Question Two

Differentiate between the following terms and give appropriate example (s) in each case

- a) Primary and Secondary air pollutants **(3 Marks)**
- b) Open dumping and sanitary landfill **(3 Marks)**

c) Facultative and maturation ponds

(4 Marks)

Question Three

a) Describe how vascular plants contribute to the treatment of wastewater and runoff in a constructed wetland.

(4 Marks)

b) Discuss how non-point sources of water pollution can be controlled

(4 Marks)

Question Four

You have been asked to be a lend expert in rectifying the problem of soil pollution in the County of Nairobi. Discuss the steps you will take to remediate polluted land.

(10 Marks)

SECTION B (30 MARKS)

Question Five

Natural precipitation is slightly acidic because of the presence of primary acidic gases in the atmosphere. However, human activities have led to an increase in the concentration of these gases in the atmosphere. Discuss in detail the impact of acid precipitation on the various aspects of the environment.

(15 Marks)

Question Six

a) Describe the various types of incinerators used to reduce the hazardous waste in the environment.

(6 Marks)

b) State two substances and their sources that deplete ozone. Discuss why the nations of the world should be concerned about ozone destruction

(9 Marks)

Question Seven

Particulate matter is an important atmospheric pollutant with several impacts on the health of the people as well as to the environment. Discuss in detail how innovated technologies work to control particulate matter from the source.

(15 Marks)
